INTRODUCTION

Prior lessons on yield measures established that a fixed-income investor's rate of return will equal a bond's yield-to-maturity (YTM) under certain assumptions. In these lessons, we explore the sources of return for fixed-income investments and demonstrate investment returns in different scenarios, including the one embedded in the YTM calculations. Prior lessons also established interest rate risk. We show how investment horizon, in relation to a bond's features, is a key determinant of interest rate risk for investors and how different investors in the same fixed-income investment can have different returns and views on risk. Finally, we introduce Macaulay duration, a weighted-average measure of the time to receipt for a bond's cash flows, and demonstrate how holding a bond for its Macaulay duration balances reinvestment and price risks.

Category
Interest Rate Risk and Return

Learning Module Overview

- There are three sources of return for fixed-rate bond investors: (1) coupon and principal payments, (2) reinvestment of coupons, and (3) gain or loss on the sale of the bond if the bond is sold prior to maturity.
- The rate of return on a fixed-rate bond investment is found by using the holding period, the future value of coupons received, the sale price, and the purchase price to calculate a compounded, annualized rate of return.
- If all cash flows are received at scheduled dates, coupons are reinvested at the same rate as a bond's YTM, and the bond is held to maturity, an investor's annualized compounded rate of return will equal the bond's YTM. If any of those assumptions do not hold, the investor's rate of return will vary.
- Reinvestment risk and price risk are types of interest rate risk and have an inverse relationship. Reinvestment risk is the risk of decreasing reinvestment returns on cash flows, which occurs when interest rates fall. Price risk refers to declining prices and occurs when interest rates rise.
- The longer the investment horizon, the more important reinvestment risk is relative to price risk. If an investor's investment horizon equals the Macaulay duration of a bond, the risks equally offset each other.
- Macaulay duration is the weighted-average time to receipt of a bond's cash flows, where the weights of each cash flow in the calculation are each cash flow's share of the bond's full price (i.e., present value).
- When the investment horizon is greater than (less than, equal to) a bond's Macaulay duration, coupon reinvestment risk is higher than (lower than, equal to) the bond's price risk.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of the learning module.

1. A source of risk for a buy-and-hold fixed-income investor, ignoring credit risk, is from:

C. neither capital gain/loss on sale nor coupon reinvestment.

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Solution:

B is correct. Changes in interest rates during the holding period of the bond will impact returns from reinvestment of coupons.

A is incorrect because for a buy-and-hold investor, there is no risk of capital gain or loss since the investor is not selling the bond prior to maturity and will get the par value of the bond at the maturity date.

C is incorrect because a buy-and-hold investor's returns are impacted by the rate earned on the reinvestment of coupons, or reinvestment risk.

2. A family office purchases a six-year, 5.8% annual coupon eurobond priced at par for settlement on 15 December 2031. The bond matures on 15 December 2037. Immediately after the purchase of the bond, interest rates rise to 6.5%. The family office sells the bond after three years. The family office's total annualized return on the investment was *closest* to:

A. 5.28%.
B. 5.80%.
C. 6.50%.

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Solution:

A is correct. The total annualized return is composed of the (1) coupon payments, (2) reinvestment of coupons, and (3) gain/loss on the sale. This is equal to 5.28%, as shown below.

The future value of the coupons including reinvestment is

= –*FV*(0.065, 3, 5.8, 0, 0) = 18.556.

The sale price of the bond after three years is

= –*PV*(0.065, 3, 5.8, 100, 0) = 98.146.

The realized return, *r*, is

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{18.556 + 98.146}{100}\right)^{\frac{1}{3}} - 1.$$

r=0.0528=5.28%

B is incorrect because it would be the return if interest rates remained at 5.8%.

C is incorrect because 6.5% is the interest rate at which the coupons will be reinvested, not the rate of return for the family office on the investment.

3. The risk to an investor who buys a bond at par value and intends to sell it before the receipt of the first coupon payment, ignoring credit risk, is *most likely* from:

A. price risk.
B. reinvestment risk.
C. neither price risk nor reinvestment risk.

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Solution:

A is correct. Changes in interest rates can affect the price of the bond and, therefore, the returns for the investor.

be reinvested.

C is incorrect because the investor faces price risk.

4. Reinvestment risk and price risk of a bond offset one another if an investor's investment horizon is:

A. equal to the bond's time-to-maturity.
B. equal to the bond's Macaulay duration.
C. less than the bond's Macaulay duration.

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Solution:

B is correct. When the investment horizon is equal to the Macaulay duration, the gain (loss) from coupon reinvestment is offset by the loss (gain) in the price of the bond from changes in interest rates.

A is incorrect because if an investor holds a bond to maturity (ignoring credit risk), the investor faces no price risk but does face reinvestment risk. Therefore, reinvestment risk will dominate price risk.

C is incorrect because if the investment horizon is less than the Macaulay duration, price risk dominates reinvestment risk.

5. Hightest Capital purchases a 15-year, 6.8% annual coupon bond and has an investment horizon of 7.0 years. The Macaulay duration of the bond is 9.85 years. The duration gap at the time of purchase is *closest* to:

A. 2.85.
B. 5.15.
C. 8.00.

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Solution:

A is correct. The duration gap for a bond is the difference between its Macaulay duration and the investor's investment horizon: 9.85 – 7.0 = 2.85.

B is incorrect because it is the difference between the bond's maturity and its Macaulay duration (15 – 9.85 = 5.15).

C is incorrect because it is the difference between the bond's maturity and the investor's investment horizon (15 – 7 = 8.00).

6. Consider a bond that has two years remaining to maturity, a coupon of 4% paid semiannually, and a yield-to-maturity of 4.60%. Assuming it is 63 days into the first coupon period and a 30/360 basis, the bond's annualized Macaulay duration is *closest* to:

A. 0.9419 years.
B. 1.7666 years.
C. 1.9416 years.

Hide Solution

Solution:

B is correct.

Period	Time to Receipt	Cash Flow	*PV*	Weight	Time to Receipt × Weight
1.0000	0.6500	2.0000	1.9707	0.0198	0.0129
2.0000	1.6500	2.0000	1.9264	0.0193	0.0319
3.0000	2.6500	2.0000	1.8830	0.0189	0.0501

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4.0000	3.6500	102.0000	93.8759	0.9420	3.4383
			99.6559	1.0000	3.5331
				Annualized MacDur	1.7666

The first cash flow's time-to-receipt is (180-63)/180 = 0.65 periods from now as it is 63 days into the period. Each subsequent cash flow is received one period later after the first, so time-to-receipt = 0.65 + 1, 0.65 + 2, and so on.

7. Consider a bond that has five years remaining to maturity, a coupon of 0% paid annually, and a yield-to-maturity of –0.38%. Assuming it is the issuance date and a 30/360 basis, the bond's annualized Macaulay duration is *closest* to:

A. 3.5361 years.
B. 5.0000 years.
C. 6.3412 years.

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Solution:

B is correct. No calculation is required, because the Macaulay duration of a zero-coupon bond is its time-to-maturity unless it is between coupon dates. To demonstrate, however, see the following table:

Period	Time to Receipt	Cash Flow	*PV*	Weight	Time to Receipt × Weight
1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
2.0000	2.0000	0.0000	0.0000	0.0000	0.0000
3.0000	3.0000	0.0000	0.0000	0.0000	0.0000
4.0000	4.0000	0.0000	0.0000	0.0000	0.0000
5.0000	5.0000	100.0000	100.9554	1.0000	5.0000
			100.9554	1.0000	5.0000

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